COMP3821/9801

Workshop Week 5

Reductions and NP*-completeness*

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Guided Problem

The demonstrators will solve the following problems. For the rest of the workshop, work in small groups to solve the remaining workshop exercises. There are optional problems at the end of this document for further practice.

Guided Problem

Consider the following pair of problems.

- IndSet: Given a graph $G = (V, E)$ and an integer $1 \leq k \leq n$, does there exist a subset $S \subseteq V$ of at least k vertices such that any pair of vertices in S are not connected by a direct edge?
- HighDegreeIndSet: Given a graph $G = (V, E)$ and an integer $1 \leq k \leq n$, does there exist a subset $S \subseteq V$ of at least k vertices such that S forms an independent set and each vertex in S has degree at least k in G?

(a) Show that HighDegreeIndSet is in NP.

(b) It is known that IndSet is NP-complete. Show that HighDegreeIndSet is NP-hard and hence, conclude that HighDegreeIndSet is NP-complete.

Guided Problem

A *balloon* of size k is an undirected graph that consists of a simple cycle of length k and a simple path of size k, where the path contains an endpoint that lies on the cycle. Every other vertex on the cycle is disjoint from every other vertex on the path. A balloon of size k has $2k$ vertices, k vertices in the cycle and $k + 1$ vertices in the path including the endpoint in the cycle.

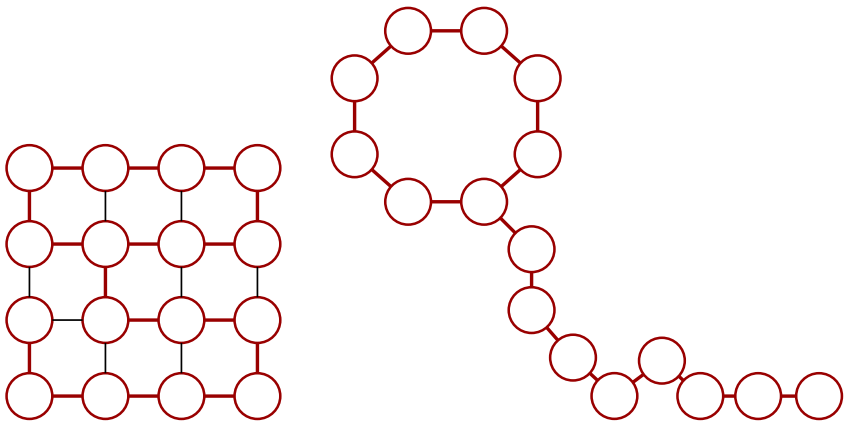


Figure 1: *A balloon subgraph of size* 8 *on the* 4×4 *grid graph.*

Given an undirected graph G and integer $1 \leq k \leq n$, prove that it is NP-complete to decide if G has a balloon of size at least k.

Workshop Exercises

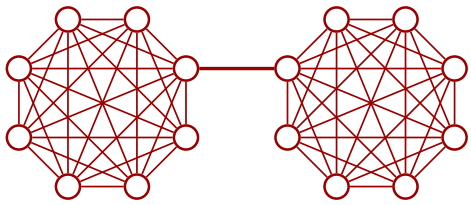
1. A *Hamiltonian cycle* in a graph G is a cycle that visits every vertices in G exactly once. It is known that deciding whether G has a Hamiltonian cycle is NP-complete.

   In this problem, we explore a slight variant of the Hamiltonian cycle problem. Here, we have a *weighted* graph. A *heavy-tonian cycle* in G is a Hamiltonian cycle whose total weight of edges is at least half of the total weight of all edges in G.

   Prove that deciding whether G contains a heavy-tonian cycle is NP-complete.

2. Let $k \geq 1$ be an integer. A k-*dumbbell* is a collection of two disjoint cliques C_1 and C_2, each of size k, together with an edge connecting C_1 and C_2.

   For example, the following is a dumbbell of size 8.

   We define the problem Dumbbell as follows: given an undirected graph $G = (V, E)$ and an integer k, decide whether there exist a k-dumbbell subgraph in G.

   Show that Dumbbell is NP-complete.

3. Consider the following problems.

   - 3Sum: Given a set[1] of n integers, does there exist three integers that sum to zero?
   - Collinear: Given a collection of points in $\mathbb{R}^2$, does there exist three points that lie on the same line?

   We say that a problem X is 3SUM-*hard* if there exists a subquadratic-time reduction from 3Sum to X. Show that Collinear is 3SUM-hard by constructing a linear-time reduction from 3Sum.

   Hint. *Suppose that $a+b+c=0$. Consider the polynomial $P(x) = (x-a)(x-b)(x-c)$.*

[1] *We require the array be a collection of distinct elements.*

Optional Exercises

These exercises will not contain solutions. We encourage that you post on the forums to discuss the approaches for these problems. Additionally, you may find the hints helpful.

4. Let $G = (V, E)$ be an undirected graph. A *Hamiltonian path* is a simple path that visits every vertex *exactly* once. It is known that deciding if a graph has a Hamiltonian path is NP-complete. Consider the related problem of deciding if there exist a simple path in G that misses 2025 vertices; in other words, we want to decide if a graph G has a simple path that visits $|V| - 2025$ vertices *exactly* once. Show that this problem is NP-complete by reducing from the Hamiltonian path problem.

   Note. *You may assume that an input graph has at least 2025 vertices; otherwise, return NO.*

5. Let $G = (V, E)$ be an undirected graph. Prove that deciding if G has a simple path that passes through *at least* half of the vertices in G is NP-complete.

6. We now generalise the previous problems. Instead of deciding if a graph contains a simple path that misses 2025 vertices, it now misses k vertices, where k is now part of the input. Prove that this problem is NP-complete.

   Note. *You can assume that k is an integer that is polynomial in the size of the graph. This just removes some of the technical subtleties in the construction of the graph, which is not the point!*

7. Let $G = (V, E)$ be an undirected graph. Consider the following concepts on G.

   - An *independent set* S of size k is a subset of k vertices such that no pair of vertices in S are connected by a direct edge.
   - A *clique* C of size k is a subset of k vertices such that every pair of vertices in C are connected by a direct edge.

   It is known that deciding if a graph has an independent set of size at least k is NP-complete. Use this to show that deciding if a graph has a clique of size at least k is also NP-complete.

8. We reuse the concepts from the previous question. Let $G = (V, E)$ be an undirected graph. A *clique-independent* set X of size k is a subset of $2k$ vertices such that:

   - X contains k vertices that form an independent set I of G,
   - X contains k vertices that form a clique C of size k, and
   - $I \cap C = \emptyset$; that is, the independent set and clique are disjoint.

   Prove that deciding if a graph has a clique-independent set of size at least k is NP-complete.

9. Let $G = (V, E)$ be an undirected graph. Prove that deciding if there exist a simple path of length at least k is NP-complete, where we measure the length of the path by the number of edges in the path.

10. Let $G = (V, E)$ be an undirected graph. Prove that deciding if G contains a spanning tree where every vertex in the spanning tree has degree at most 2025 is NP-complete. As an exercise, generalise this result!

11. Let $G = (V, E)$ be an undirected graph. A *proper 3-colouring* of G is an assignment of vertices to *at most* three colours so that any pair of adjacent vertices do not share the same colour. It is known that deciding if a graph has a proper 3-colouring is NP-complete. Use this to show that deciding if a graph has a proper 4-colouring is NP-complete.

    Note. *A proper 4-colouring is an assignment of vertices to at most four colours with the same condition as a 3-colouring.*

12. Use the same idea as above to prove that deciding if a graph has a proper k-colouring is NP-complete for $k \geq 3$.

13. Let A be an array of n integers. We consider the following two problems on A.

    - SubsetSum: Given an array A of n integers and an integer T, decide if there exist a subset of integers in A that sum to T.
    - Partition: Given an array A of n integers, decide if there exist a way to partition A into two sets E_1 and E_2 such that
    $$\sum_{i \in E_1} A[i] = \sum_{i \in E_2} A[i].$$

    It is known that Partition is NP-complete. Use this to show that SubsetSum is also NP-complete. See Problem 30 for another proof of the NP-completeness of SubsetSum.

14. Consider the following problems.

    - HamiltonianCycle: Given an undirected and unweighted graph $G = (V, E)$, decide if there exist a simple cycle in G that visits every vertex exactly once.
    - TravellingSalesperson: Given a complete and weighted graph G and an integer k, decide if there exist a simple cycle in G that visits every vertex exactly once with total weight at most k.

    It is known that HamiltonianCycle is NP-complete. Use this to show that TravellingSalesperson is also NP-complete.

    Note. *The input to the Travelling Salesperson problem is a complete graph, not an arbitrary graph.*

15. A *kite subgraph* of size k is an undirected graph that consists of a clique of size k and a simple path of size k, where the path contains an endpoint that lies on the clique. Every other vertex on the clique is disjoint from every other vertex on the path. A balloon of size k has $2k$ vertices, k vertices in the cycle and $k + 1$ vertices in the path including the endpoint in the cycle.

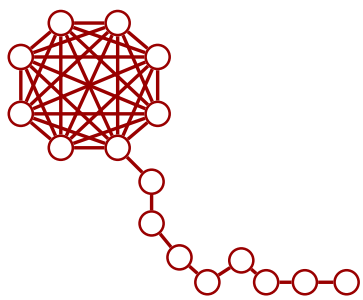


Figure 2: *A kite subgraph of size* 8.

Given an undirected graph G and an integer k, prove that it is NP-complete to decide if G has a kite subgraph of size at least k.

16. A *dandelion* of length k is an undirected graph that consists of a simple path of length k, followed by k additional vertices connected to one of the endpoints.

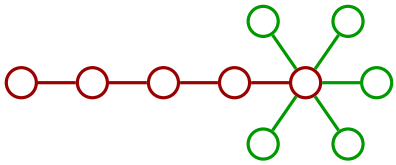
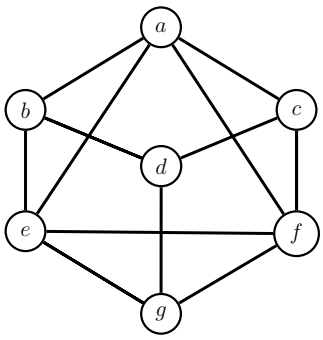


Figure 3: *A dandelion of length* 5.

Given a graph G and integer $k \geq 1$, prove that it is NP-complete to decide if G has a dandelion subgraph of length at least k.

17. A *doubly Hamiltonian tour* is a closed walk in an undirected graph G that visits every vertex exactly twice. For example, the following graph

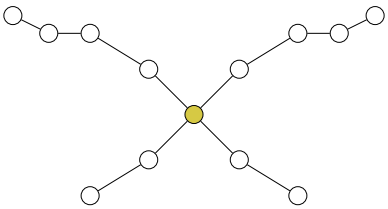


has the doubly Hamiltonian tour

$$a \to b \to d \to g \to e \to b \to d \to c \to f \to a \to c \to f \to g \to e \to a.$$

Given an undirected graph G, prove that deciding whether G contains a doubly Hamiltonian tour is NP-complete.

18. Let $G = (V, E)$ be an undirected graph. A *spanning spider* $S = (V, E')$ is a spanning tree of G such that there is at most one vertex with degree strictly larger than[2] 2 in the induced subgraph $G[S]$; that is, there is at most one vertex of degree at least 2 in the graph generated by removing edges not in E'.



A spanning spider where the vertex of degree 2 or more is highlighted.

Prove that deciding if G has a spanning spider is NP-complete.

19. Let $G = (V, E)$ be an undirected graph. Prove that deciding if G has a spanning tree with at most 2025 leaves is NP-complete.

[2] *Otherwise, all vertices in the legs of the spider would make the spider infeasible.*

20. Consider the following problem.

    HalfClique

    Input. An undirected graph $G = (V, E)$ on $2n$ vertices.

    Task. Decide if G has a clique of size n.

    Show that HalfClique is NP-complete.

21. Consider the following problem.

    CliquePartition

    Input. An undirected graph $G = (V, E)$ and an integer k.

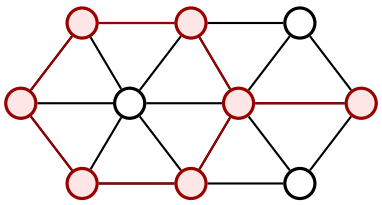
    Task. Decide if there exist a partition of V into sets $V_1, \ldots, V_k$ such that for each $1 \leq i \leq k$, V_i forms a clique.

    Prove that CliquePartition is NP-complete.

22. Recall that in Question 13 we had a reduction from Partition to SubsetSum. Prove that Partition is NP-complete by reducing from SubsetSum.

    Note. *There is a reason why this is a lot later in the problem set! This direction is much harder. Here, we need to relax the set condition because we aren't guaranteed that the elements in the subset sum are distinct, so you may assume that a subset can also contain duplicate elements as well.*

23. Let $G = (V, E)$ be an undirected and unweighted graph. A subset $S \subseteq V$ of vertices is $\square$*-free* if, for every four vertices $u, v, w, x \in S$, at least one of $(u, v), (v, w), (w, x), (u, x)$ does not appear in the edge set of G. In other words, G induced by S has no cycle of length 4. Prove that deciding if G has a $\square$-free subset of k vertices is NP-complete.



*A $\square$-free subset of 7 vertices with its induced edges. Note that this is **not** the largest subset.*

24. We consider the variant of the clique problem as follows:

    PartitionedClique

    Input. A graph $G = (V, E)$, where V is partitioned into k sets $V_1, \ldots, V_k$.

    Task. Decide if G has a clique of size k, containing one vertex from each set V_i.

    Prove that PartitionedClique is NP-complete.

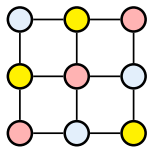
25. Let $G = (V, E)$ be an undirected graph. A *poor colouring* of G assigns each vertex a set of two *distinct* colours such that each edge must use two different sets of colours. However, they may share a common colour. In other words, the endpoints of each edge in a poor colouring can share *at most* one common colour. Call a poor colouring that uses at most three colours a *poor 3-colouring*.

    Prove that deciding if G has a poor 3-colouring is NP-complete.

26. Let $G = (V, E)$ be an undirected graph. A *rich colouring* assigns each vertex a set of two colours such that each edge must use two different sets of colours. Additionally, they may not share any colour. In other words, the endpoints of each edge in a rich colouring uses *exactly* four distinct colours. Therefore, every rich colouring is also a poor colouring. Call a rich colouring that uses at most five colours a *rich 5-colouring*.

    Prove that deciding if G has a rich 5-colouring is NP-complete.

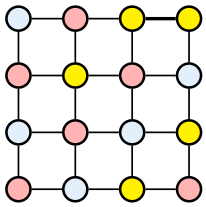
27. Let $G = (V, E)$ be an undirected graph. Recall that a proper 3-colouring is a colouring of the vertices of G using at most three colours, so that no pair of adjacent vertices share the same colour. We explore a slightly weaker variant, which we call Balanced3Col. In this problem, we are still given an undirected graph G but this time, we colour the vertices such that each colour has the same number of vertices. In a balanced 3-colouring, we still require any pair of adjacent vertices to not share the same colour.

    Prove that Balanced3Col is still NP-complete.

28. Let G be an undirected graph. Recall that a proper 3-colouring is a colouring of the vertices of G using at most three colours, so that no pair of adjacent vertices share the same colour.

    In this problem, we explore a slightly weaker variant, which we call Nearly3Col. In this problem, we are still given an undirected graph G but this time, we allow *at most* one edge's endpoints in G to have the same colour.

    Prove that Nearly3Col is still NP-complete.

29. Consider the following problem.

    3DimensionalMatching

    Input. Three sets X, Y, Z, each of size n, and a set $T \subset X \times Y \times Z$ of ordered triplets; that is, T is composed of triplets (x, y, z) such that $x \in X, y \in Y, z \in Z$.

    Task. Decide if there exist a set $S \subseteq T$ of n triplets in T such that each element of X, Y, Z is contained in *exactly* one triplet in S.

    Prove that 3DimensionalMatching is NP-complete.

    Note. *In some sense, the two-dimensional variant can be thought of as deciding if a bipartite graph has a perfect matching.*

30. Recall the SubsetSum problem from Question 13. Show that it is NP-complete by reducing from 3DimensionalMatching.

31. A sequence of points $S_i = (x_i, y_i)$, for $i = 1$ to k is a *convex polygon* (in counter-clockwise order) if, in the sequence T of size $k + 2$ consisting $S_1, \ldots, S_k, S_1, S_2$, each adjacent triple of points T_i, T_{i+1}, T_{i+2} represents a left turn, for $i = 1$ to k. The *convex hull* of a point set P is a subset S of P such that S is a convex polygon and every point of $P \setminus S$ is interior to S. We now consider the following two problems.

    - SORTING: Given a set of n integers, output the sequence of integers in increasing order.
    - CONVEXHULL: Given a sequence S of n points in $\mathbb{R}^2$, compute the sequence of points that lie on the convex hull of S.

    Show that there exists a linear-time reduction from SORTING to CONVEXHULL.

    Note. *This implies that any counter-clockwise based algorithm for convex hull requires* $\Omega(n \log n)$ *time.*

Hints

4. Try adding a few vertices in G to form G', so that if G has a Hamiltonian path, then the same path must miss the vertices you added in G'. How many vertices should you add?
5. Imitate the previous argument, but you want to add some number of vertices in terms of the number of vertices in the input graph.
6. Generalise the above arguments.
7. Try proving that an independent set of G corresponds to a clique of $\overline{G}$, where $\overline{G}$ is the complement graph. The converse is true as well!
8. The easiest reduction reduces from the clique problem. Try adding a few vertices to form an independent set.
9. A graph has a simple path of length $n-1$ if and only if the graph has a Hamiltonian path.
10. Reduce from the Hamiltonian path problem. Any spanning tree where every vertex has degree at most 2 in the tree is a Hamiltonian path! Try blowing up the degree of the vertices and repeat the same ideas as the previous problem.
11. If G has a proper 3-colouring, add a few vertices to force any such colouring to use the additional colour.
12. Generalise the above argument.
13. Let E_1 and E_2 be a partition of A, so that the sums are the same. Choose a particular value for T so that either E_1 or E_2 form a subset sum that sum to T.
14. It is easy to generate a complete graph from the input graph G. But you need to give weights to the edges of the complete graph appropriately, so that any Hamiltonian cycle has weight at most k, where you can freely choose k.
15. Reduce from the clique problem.
16. Reduce from the Hamiltonian path problem; take inspiration from the earlier argument.
17. Reduce from the Hamiltonian cycle problem.
18. Reduce from the Hamiltonian path problem; make two copies of the graph. When does a spanning spider exist in the graph you constructed?
19. Reduce from the Hamiltonian path problem.
20. Reduce from the clique problem. If G has a clique of size k, add n vertices and find a way to extend this to a clique of size n.
21. You can choose the particular value for k. Consider the complement graph. Every clique of G is an independent set of $\overline{G}$, and each independent set also forms a colour class in a proper colouring. Therefore, reduce from the 3-colouring problem. Choose a specific value for k.
22. Add a particular element x into the array. To figure out what element to add, consider what happens when a partition exists. If E_1, E_2 form a partition of A so that their sums are equal, we know that

$$\sum_{i \in E_1} A[i] = \sum_{i \in E_2} A[i], \ \sum_{i \in E_1} A[i] + \sum_{i \in E_2} A[i] = \sum_{i=1}^{n} A[i].$$

One of E_1 or E_2 must contain the element x. Find an expression for x.

23. Reduce from the independent set problem.
24. Reduce from the clique problem; a pair of vertices in a clique might belong to the same vertex set. Make a few copies of the same vertex, where the copies all belong to different vertex sets.
25. Reduce from the 3-colouring problem.
26. Reduce from the 3-colouring problem; throw away two of the colours by introducing a new vertex that is connected to every vertex. Subdivide each edge by a path of length 3.
27. Reduce from the 3-colouring problem; make three copies of the graph. Argue why the new graph has a balanced 3-colouring if and only if the original graph has a proper 3-colouring.
28. Construct a gadget to describe the property that in a nearly 3-colouring of X, exactly one edge of X has endpoints that have the same colour.
29. Reduce from 3SAT; encode each variable x_i as a gadget composed of a triangle. Think about how to construct clause gadgets and finally, how we connect clause and variable gadgets.
30. Associate each edge with a $3n$-digit number.
31. Map each integer to a point in $\mathbb{R}^2$ such that a sorted array corresponds to a counter-clockwise sequence of points on the convex hull.